

**Workshop on Computational Models of Human Road User Behavior for Evaluation of Automated Vehicles**

**September 15, 2026, Naples, Stazione Marittima, Room Calipso**

Abstracts of contributed talks and most recent schedule are available at <https://roadusermodels.github.io/>

| Session                               | Start | End   | Duration<br>(min) | Item Type   | Speaker(s)                     | Affiliation                                | Presentation Title / Topic                                                                                                        |
|---------------------------------------|-------|-------|-------------------|-------------|--------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Session 1.<br>Chair: Arkady Zgonnikov | 8:30  | 8:40  | 10                | Logistics   | Arkady Zgonnikov               | TU Delft, Organizer                        | Welcome, Workshop Objectives & Agenda                                                                                             |
|                                       | 8:40  | 9:10  | 30                | Keynote     | Biagio Ciuffo                  | European Commission – JRC                  | Driving behaviour models to operationalize the international regulatory framework for AV safety approval                          |
|                                       | 9:10  | 9:40  | 30                | Keynote     | Hiroki Nakamura & Sou Kitajima | Japan Automobile Research Institute (JARI) | Defining Safety Boundaries for Automated Driving Systems: From real traffic data to computational human road-user models          |
| Session 2.<br>Chair: Jonas Bärghman   | 9:40  | 10:00 | 20                | Contributed | Julian Frederik Schumann       | TU Delft                                   | Active Inference as a model of human driving behavior                                                                             |
|                                       | 10:00 | 10:20 | 20                | Contributed | Arturo Tejada Ruiz             | TNO                                        | Bending the (traffic) rules, competent driver style!                                                                              |
|                                       | 10:20 | 10:40 | 20                | Contributed | Giovanni Albano                | Univ. of Naples Federico II                | Risk perception and braking reaction model of competent and careful human drivers in safety-critical cut-in scenarios             |
|                                       | 10:40 | 11:00 | 20                | Contributed | Federico Scari                 | RDW                                        | A Scenario-Based Grading Framework for ADS/ADAS Consumer-Testing Campaigns Benchmarked Against a Probabilistic Human Driver Model |
| Break                                 | 11:00 | 11:30 | 30                |             |                                | Coffee Break                               |                                                                                                                                   |
| Session 3.<br>Chair: Gustav Markkula  | 11:30 | 12:00 | 30                | Keynote     | Jonas Bärghman                 | Chalmers Univ. of Technology               | Representativeness in ADS safety impact assessment (and training): baseline and traffic-agent challenges                          |
|                                       | 12:00 | 12:30 | 30                | Keynote     | Manel Hammouda                 | BMW Group                                  | Cognition and Reward: A Dual Perspective on Driver Behavior Modeling                                                              |
|                                       | 12:30 | 12:50 | 20                | Contributed | Yueyang Wang                   | Univ. of Leeds                             | Task-oriented scoring of driver behaviour models for automated driving systems evaluation                                         |
|                                       | 12:50 | 12:53 | 3                 | Teaser talk | Malin Svärd                    | Volvo Cars                                 | Quantification of Human Acceleration Behavior across Speeds Using Real-World Driving Data                                         |
|                                       | 12:53 | 12:56 | 3                 | Teaser talk | Konstantinos Mattas            | European Commission – JRC                  | Combining Evasive and Braking Reactions for Safety Reference Models in Automated Vehicles                                         |
|                                       | 12:56 | 12:59 | 3                 | Teaser talk | Samir Mohammad                 | TU Delft                                   | General-purpose LLMs as Models of Human Driver Behavior: The Case of Simplified Merging                                           |
| Break                                 | 13:00 | 14:00 | 60                |             |                                | Lunch Break                                |                                                                                                                                   |
| Session 4.<br>Chair: Luigi Pariota    | 14:00 | 14:30 | 30                | Keynote     | Gustav Markkula                | Univ. of Leeds                             | Mechanistic, machine-learning, and hybrid modelling of road user behaviour                                                        |
|                                       | 14:30 | 14:50 | 30                | Keynote     | Johan Engström                 | Waymo                                      | SAE J3330: Toward a shared language for driving behavior reference models                                                         |
| Break                                 | 15:00 | 15:15 | 15                |             |                                | Break                                      |                                                                                                                                   |
| Session 5.<br>Chair: Johan Engström   | 15:15 | 15:35 | 20                | Contributed | Arash Tavakoli                 | Villanova Univ.                            | Driver World Models: Unified Representations of Drivers in Semi-Automated Driving                                                 |
|                                       | 15:35 | 15:55 | 20                | Contributed | Christian Rössert              | cogniBIT GmbH                              | Beyond benchmark metrics: a cognitive driver model as a behavioral reference for AV evaluation                                    |
|                                       | 15:55 | 16:15 | 20                | Contributed | Kevin Kefauver                 | VTTI                                       | Prudent Attentive Sober (PAS) Driver Definition in US National Crash Data and its Application to Human Behavioral Models          |
|                                       | 16:15 | 16:35 | 20                | Contributed | Saeed Rahmani                  | TU Delft                                   | Drive with Style: Style-Conditioned Self-Play for Behaviorally Diverse Traffic Simulation                                         |
| Break                                 | 16:35 | 16:45 | 10                |             |                                | Break                                      |                                                                                                                                   |
| Session 6.<br>Chair: Shu-Yuan Liu     | 16:45 | 17:35 | 50                | Discussion  | Shu-Yuan Liu, Arkady Zgonnikov | Moderated by Organizers                    | Interactive Breakout Groups & Plenary Synthesis                                                                                   |
|                                       | 17:35 | 17:45 | 10                | Logistics   | Arkady Zgonnikov               | Organizing Committee                       | Closing Remarks                                                                                                                   |

| 30-50 minutes                                                                                            |                                                                                                                                |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|
| <b>Total attendees – 30-50 ppl; expected: 40-50</b>                                                      |                                                                                                                                |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| Time                                                                                                     | Duration                                                                                                                       | Task                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 16:45-16:50                                                                                              | 5 minutes                                                                                                                      | Objective setting and group formation                                                                                           | Instructions: form groups of 4-5 ppl, ideally with someone you don't know.<br>1) Round of introductions (30 seconds each person)<br>2) If there is time: Icebreaker: Tell the group about something you're good at (not research/work-related)<br>3) Choose the topic, agree on your version of the answer to the big question.<br>4) Write down your answer (if <=4 groups) or type it in the web form (if >4 groups)<br>5) If you finalized an answer but there is time left, pick up another topic.                                                                                                                                                                                                                                                              |  |  |  |  |  |  |
| 16:50-17:00                                                                                              | 30 minutes                                                                                                                     | Group discussion                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 17:00-17:30                                                                                              | 10 minutes                                                                                                                     | Plenary session                                                                                                                 | If 8 groups or less: Each group has 1 minute to give their answer(s) and briefly explain. If >8 groups: type the answers in a web form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  |  |  |  |
| 17:30-17:35                                                                                              | 5 minutes                                                                                                                      | Summary                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| Track                                                                                                    | Moderators/Leads                                                                                                               | The big question                                                                                                                | Examples of directions for discussion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |  |  |  |  |  |  |
| Creating models of road user behavior                                                                    |                                                                                                                                | In the future, what would it take modelers to convincingly demonstrate that a model of human behavior is the right one?         | -Mechanistic, data-driven, or hybrid – what does each actually buy us, and what does "hybrid" mean concretely beyond the slogan?<br>-Is interpretability a genuine requirement, or a proxy for trust that a sufficiently well-validated black box would make unnecessary?<br>-What data, metrics, or know-how is missing for "ideal" validation, and which of those gaps is closest to being closed?<br>-How do we validate closed-loop behavior, where the model changes the situation it is responding to and there is no ground truth?<br>-Does "right" mean the same thing for a model built to explain behavior and a model built to generate it?                                                                                                              |  |  |  |  |  |  |
|                                                                                                          | No centralized moderators; if we see that some groups consist mainly of speakers, we will ask them to spread over other groups | Which important ingredient(s) is missing from current definitions of "careful and competent human driver"?                      | - Which population/data defines competence: crash-data subsets, professional drivers, naturalistic fleet data?<br>- Are CAC reference models mostly useful for collision/conflict avoidance, or also in routine driving scenarios?<br>- How do we reconcile the regulator's need for simple, foolproof reference models with the fact that human behavior is very complex?<br>- Once the number of reference models used in practice inevitably increases, how should model users navigate this landscape?<br>- Can models of careful and competent drivers be adapted into models of other sub-populations needed to realistically represent surrounding traffic (distracted, impaired, aggressive drivers)? Or does this field need principally different models? |  |  |  |  |  |  |
| The use of human driver models as behavior references for AVs                                            |                                                                                                                                |                                                                                                                                 | -What other applications can benefit from road user modeling and how? Driver assistance system design, training data generation, driver monitoring, personalization?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |  |  |  |
| Beyond "careful and competent driving": human behavior models for traffic simulation and other use cases |                                                                                                                                | Which other application of road user modeling is most promising today? What would it take to achieve it?                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| <b>Total attendees &gt; 30</b>                                                                           |                                                                                                                                |                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| Time                                                                                                     | Duration                                                                                                                       | Task                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 16:45-16:50                                                                                              | 5 minutes                                                                                                                      | Objective setting and group formation                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 16:50-16:58                                                                                              | 8 minutes                                                                                                                      | Subgroup discussion                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 16:58-17:00                                                                                              | 2 minutes                                                                                                                      | Group synthesis                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 17:00-17:30                                                                                              | 30 minutes                                                                                                                     | Group presentation and QA (10 minutes each group)                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| 17:30-17:35                                                                                              | 5 minutes                                                                                                                      | Summary                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |  |  |  |  |  |
| Track                                                                                                    | Moderators/Leads                                                                                                               | Key questions (2 max)                                                                                                           | Suggested outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |  |  |  |  |  |
| Regulation and standardization                                                                           | Biagio Chiffi/Johan Engström                                                                                                   | 1. How do we operationalize concepts like "careful and competent drivers" and get regulatory approval?                          | Top 3 regulatory blockers for model evaluations from regulatory perspectives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |  |
|                                                                                                          | Kevin Kefauver                                                                                                                 | 2. What are the key gaps in SAE J3330? And how can the international modeling community help?                                   | Top 3 priorities in standardization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |  |  |  |
| Safety evaluation of ADS                                                                                 | Jonas Bårgman                                                                                                                  | 1. How to create benchmarking dataset with naturalistic driving data that is statistically representative?                      | Top 3 criteria for benchmarking datasets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |  |  |  |  |
|                                                                                                          | Yueyang or Samir?                                                                                                              | 2. How do we incorporate simple and longstanding kinematic metrics such as TTC, PET into more comprehensive scoring frameworks? | Top 3 considerations for forming scoring metrics or frameworks for ADS evaluations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |  |  |  |  |  |
| Behavioral modeling                                                                                      | Gustav Markkula                                                                                                                | 1. Where do data-driven methods succeed? Where do mechanistic models succeed?                                                   | Top 3 recommendations for utilizing data-driven and mechanistic models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |  |  |  |
|                                                                                                          | Julian or Saeed?                                                                                                               | 2. Do we need regulations or standardizations on driver models?                                                                 | Top 3 regulatory blockers for behavioral modeling from modeller perspectives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |  |  |  |

[illegible]